

Unit – 3 : Session – 4 : SLO – 2 | FCFS and SJF Programs — Missing C Code (SJF Non-Preemptive)

Answers

Condition: $bt[i] > bt[j]$ (swap when the earlier job is longer, so the shortest burst comes first).

Waiting time: $wt[i] = wt[i-1] + bt[i-1]$;

Turnaround time: $tat[i] = wt[i] + bt[i]$;

Completed program

```
#include <stdio.h>
int main() {
    int n, i, j;
    int bt[20], p[20], wt[20], tat[20], temp;

    float avg_wt = 0, avg_tat = 0;
    printf("Enter number of processes: ");
    scanf("%d", &n);
    printf("Enter burst time for each process:\n");
    for(i = 0; i < n; i++) {
        printf("P[%d]: ", i+1);
        scanf("%d", &bt[i]);
        p[i] = i + 1;
    }
    for(i = 0; i < n - 1; i++) {
        for(j = i + 1; j < n; j++) {
            if(bt[i] > bt[j]) { // condition: shorter burst first
                temp = bt[i]; bt[i] = bt[j]; bt[j] = temp;
                temp = p[i]; p[i] = p[j]; p[j] = temp;
            }
        }
    }
    wt[0] = 0;
    for(i = 1; i < n; i++) {
        wt[i] = wt[i-1] + bt[i-1]; // prev wait + prev burst
    }
    for(i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i]; // waiting + burst
        avg_wt += wt[i];
        avg_tat += tat[i];
    }
    printf("\nProcess\tBurst Time\tWaiting Time\tTurnaround Time\n");
    for(i = 0; i < n; i++) {
        printf("P[%d]\t%d\t%d\t%d\n", p[i], bt[i], wt[i], tat[i]);
    }
    printf("\nAverage Waiting Time = %.2f", avg_wt/n);
    printf("\nAverage Turnaround Time = %.2f", avg_tat/n);
    return 0;
}
```